

Faculty of Engineering and Technology.

Department of Electrical and Computer Engineering.

Machine Learning and Data Science - ENCS5341

Data Preprocessing & Exploratory Data Analysis (EDA)

Assignment No.1

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Section:2

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❖ Data Loading and Initial Inspection

In this section, the dataset was loaded into a **Pandas DataFrame** using *pd.read_csv("customer_data.csv")* to enable easy data manipulation and analysis. An initial inspection was conducted to understand the structure and content of the data:

❖ The **. head()** method returns the first 5 rows if a number is not specified.

```
file.head()
```

	CustomerID	Age	Gender	Income	Tenure	ProductType	SupportCalls	ChurnStatus
0	CUST0000	59.0	0	151203.0	4.0	0	1.0	0
1	CUST0001	69.0	0	58332.0	6.0	1	9.0	0
2	CUST0002	46.0	1	149481.0	2.0	0	12.0	0
3	CUST0003	32.0	1	115937.0	1.0	1	13.0	0
4	CUST0004	60.0	0	103929.0	4.0	1	5.0	0

Figure 1:head() method

❖ The **.info()** method prints information about the DataFrame. The information contains the number of columns, column labels, column data types, memory usage, range index, and the number of cells in each column (non-null values).

```
file.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 3500 entries, 0 to 3499
Data columns (total 8 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   CustomerID    3500 non-null   object
 1   Age           3325 non-null   float64
 2   Gender        3500 non-null   int64
 3   Income        3328 non-null   float64
 4   Tenure        3325 non-null   float64
 5   ProductType   3500 non-null   int64
 6   SupportCalls  3329 non-null   float64
 7   ChurnStatus   3500 non-null   int64
dtypes: float64(4), int64(3), object(1)
memory usage: 218.9+ KB
```

Figure 2:info() method.

❖ The **.describe()** method returns description of the data in the DataFrame.
If the DataFrame contains numerical data, the description contains these information for each column: count - The number of not-empty values,mean - The average (mean) value,std - The standard deviation,min - the minimum value,25% - The 25% percentile,50% - The 50% percentile,75% - The 75% percentile,max - the maximum value.

```
[6]: file.describe()
```

[6]:

	Age	Gender	Income	Tenure	ProductType	SupportCalls	ChurnStatus
count	3325.000000	3500.000000	3.328000e+03	3325.000000	3500.000000	3329.000000	3500.000000
mean	43.606617	0.495714	1.406860e+05	5.041504	0.298857	10.078702	0.044857
std	14.931487	0.500053	4.333271e+05	2.571029	0.457822	21.735374	0.207020
min	14.000000	0.000000	2.503700e+04	0.000000	0.000000	1.000000	0.000000
25%	31.000000	0.000000	5.653025e+04	3.000000	0.000000	3.000000	0.000000
50%	43.000000	0.000000	8.953250e+04	5.000000	0.000000	7.000000	0.000000
75%	56.000000	1.000000	1.215025e+05	7.000000	1.000000	11.000000	0.000000
max	69.000000	1.000000	5.004849e+06	9.000000	1.000000	200.000000	1.000000

Figure 3:describe() method.

❖ Handling Missing Data

First, missing values were identified using the **.isnull().sum()** function to check how many values were missing in each column.

```
file.isnull().sum()

CustomerID        0
Age             175
Gender            0
Income          172
Tenure          175
ProductType       0
SupportCalls    171
ChurnStatus       0
dtype: int64
```

Figure 4:isnull().sum() method

After inspecting the results, some numerical features such as **Age**, **Income**, **Tenure**, and **SupportCalls** contained missing values.

To handle the missing values we have two options:

1-Deleting the row with missing data: The analysis showed that approximately **%5** of the data was missing. Removing **%5** of data may **not** be a good idea. If we lose %5 of our sample size, it may reduce the statistical power and the efficiency of our model.

2-Imputation: To handle these missing values, imputation **was chosen** instead of deletion, since removing data could lead to loss of valuable information**.**

Before deciding which imputation method to use, the **distribution** of the data was analyzed using **skewness.**

```
Skewness of Age: -0.012
Skewness of Income: 9.899
Skewness of Tenure: -0.034
Skewness of SupportCalls: 7.199
```

- The skewness of **Age** and **Tenure** are almost zero which they have normally distributed, later on we will see that **Age** and **Tenure** have **no** outliers so the **mean** value was considered appropriate for these columns since the mean is sensitive to outliers.
- However, **Income** and **SupportCalls** have **positive skew** and contained **outliers**, the **median** was chosen as a more robust measure, so the **median** value was considered appropriate for these columns since the **median** is not sensitive to outliers.

  -Note: mean and median are suitable for numerical values while mode is suitable for categorical values.

```
file['Age'] = file['Age'].fillna(file['Age'].mean())
file['Income'] = file['Income'].fillna(file['Income'].median())
file['Tenure'] = file['Tenure'].fillna(file['Tenure'].mean())
file['SupportCalls'] = file['SupportCalls'].fillna(file['SupportCalls'].median())
```

Figure 5:handling missing data

```
After handling missing data
CustomerID      0
Age             0
Gender          0
Income          0
Tenure          0
ProductType     0
SupportCalls    0
ChurnStatus     0
dtype: int64
```

❖ Handling Outliers

In this section, outliers were identified using the **Z-score method** applied to the numerical features (*Age*, *Income*, *Tenure*, and *SupportCalls*). This method is used to measures how far a data point is from the mean. Formula: z = (X — mean)/standard deviation. A threshold of |Z| > 3 is often used to detect outliers.

```
from scipy import stats
numeric_cols = ['Age', 'Income', 'Tenure', 'SupportCalls']
z_scores = stats.zscore(file[numeric_cols])
abs_z_scores = abs(z_scores)
outliers = (abs_z_scores > 3)
```

(a)

```
Column: Age
Number of outliers: 0
Outlier values:
[]

Column: Income
Number of outliers: 50
Outlier values:
[5003294. 2000000. 2000000. 2000000. 5003864. 2000000. 5002990. 2000000.
 5000000. 2000000. 2000000. 2000000. 5000000. 2000000. 5000000. 2000000.
 2000000. 2000000. 2000000. 5000000. 5000000. 2000000. 5000000. 2000000.
 2000000. 5000000. 5000000. 2000000. 2000000. 5004692. 2000000. 2000000.
 5000000. 5004849. 2000000. 5003437. 2000000. 2000000. 2000000. 2000000.
 2000000. 5000000. 2000000. 5000000. 5000000. 2000000. 5003663. 5000000.
 2000000. 5000000.]

Column: Tenure
Number of outliers: 0
Outlier values:
[]

Column: SupportCalls
Number of outliers: 70
Outlier values:
[100. 150. 100. 150. 200. 150. 100. 150. 200. 100. 100. 200. 200. 100.
 150. 200. 150. 100. 150. 200. 200. 100. 200. 200. 200. 200. 200. 100.
 100. 150. 150. 150. 150. 100. 100. 150. 150. 200. 150. 150. 200. 100.
 150. 200. 200. 100. 100. 200. 100. 200. 150. 100. 150. 100. 100. 200.
 150. 150. 150. 150. 100. 150. 200. 100. 150. 200. 200. 100. 100. 200.]
```

Figure 6:Identify outliers

- **Strategies to handle outliers:**

  Before we do any treatment ,we first need to ascertain whether the outlier is data entry error or it is a genuine value.

  1-Remove the rows containing outliers value: Use this method if the number of outliers in overall data is limited or when they are caused by data entry errors or measurement mistakes, but this would not be done since we have approximately **%2** of data which contains outliers.

  2-Replace the outliers values with central tendency e.g. median: Because median not affected by outliers.

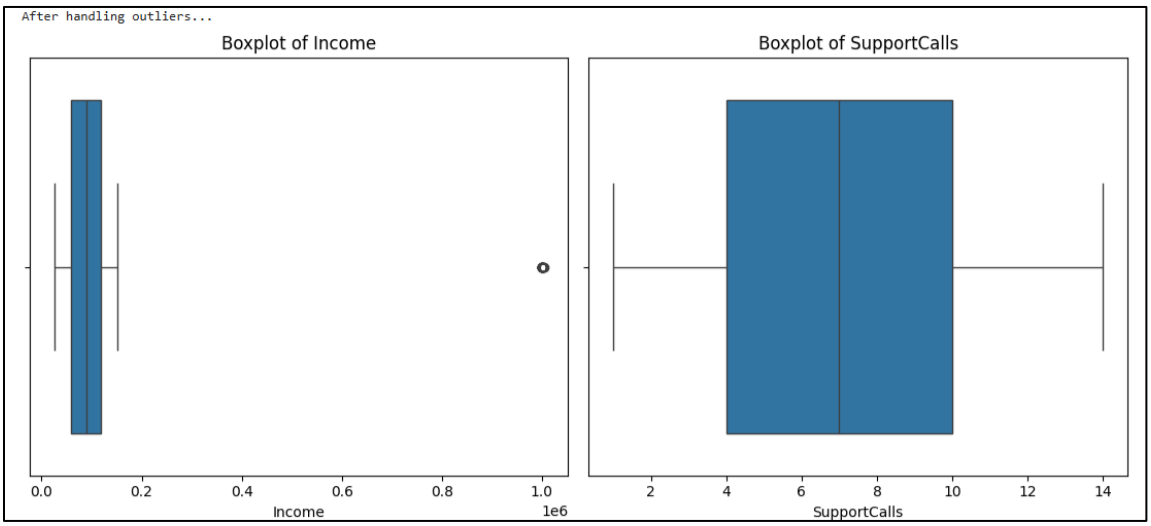


Figure 7:handle outliers using median

3-Try appropriate transformation (e.g. Log Transformation): This method does **not guarantee** elimination of outliers but it reduces the influence of those outliers values and **reduces** skewness in the data.

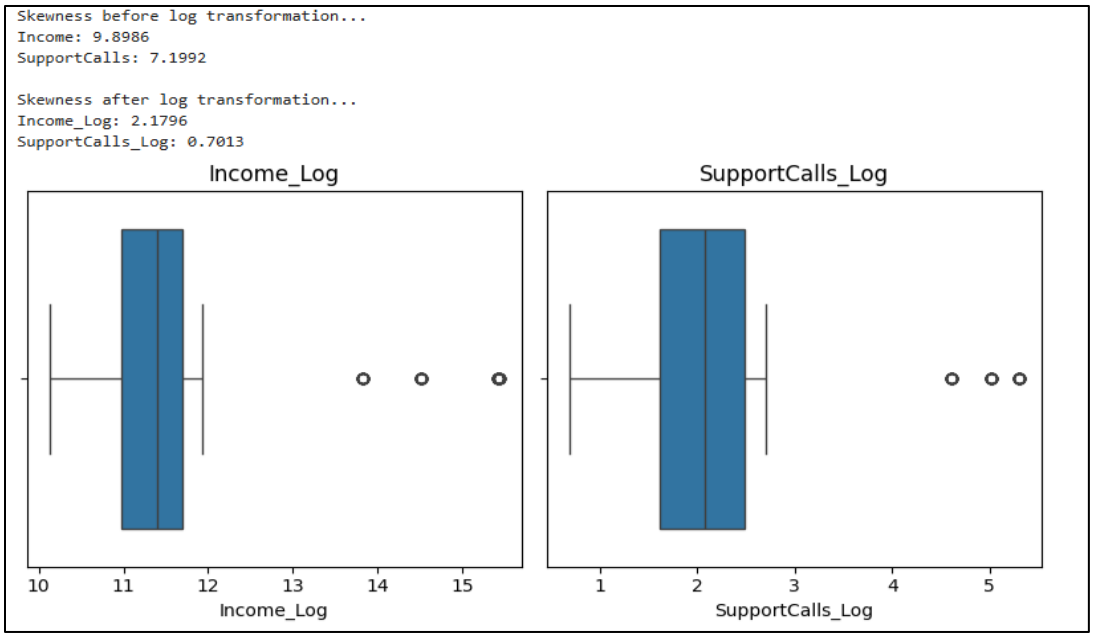


Figure 8:handle outliers using log transformation.

4-Winsorization (Capping): Winsorization is a way of handling the impact of extreme values or outliers. The statistical method involves capping the data and replacing the most extreme points—those that don't fit the rest of the data—with less extreme values from the same dataset.

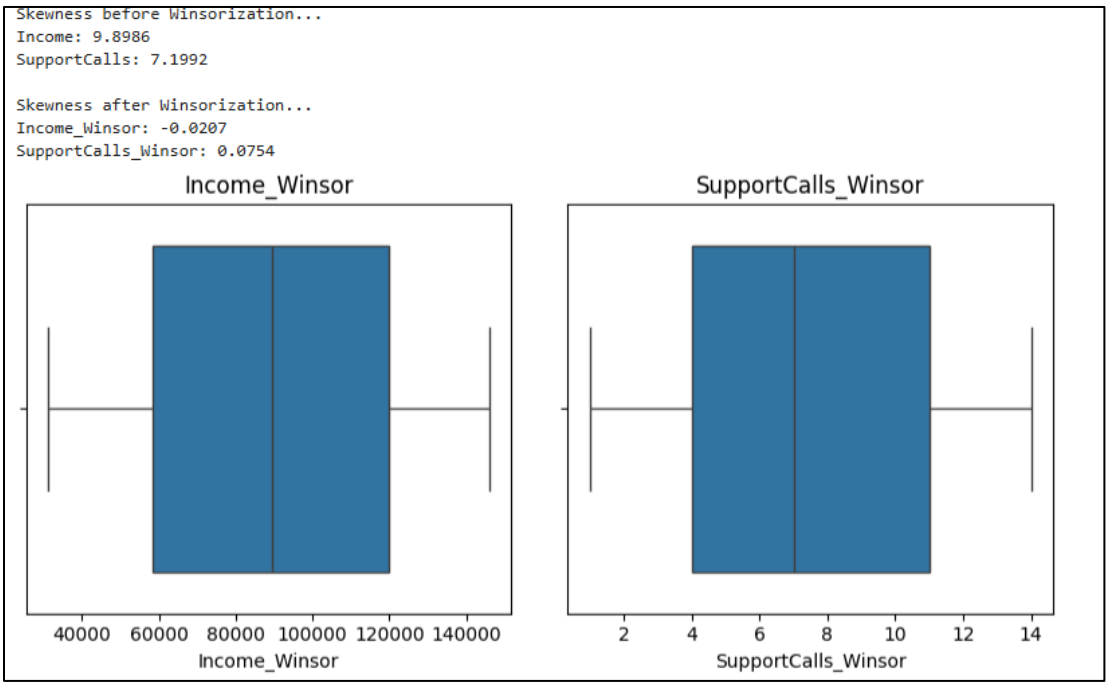


Figure 9:handle outliers using winsorization.

So, the most effective way is handling outliers using **winsorization (Capping)** method because it reduces the influence of outliers, make the distribution normal and remove them completely.

❖ Feature Scaling

In this section, we use **Standardization** that transforms data to have a mean of 0 and a standard deviation of 1.The data has an almost normal distribution and outliers are handled so the best choice is standardization (Z-score Scaling). The formula for Standardization is:

$$X' = \frac{X - \mu}{\sigma}$$

```
Checking Standardization ...
Age                    5.603137e-16
Income_Winsor          7.917476e-17
SupportCalls_Winsor    2.639159e-17
Tenure                -1.192697e-16
dtype: float64
Age                    1.000143
Income_Winsor          1.000143
SupportCalls_Winsor    1.000143
Tenure                 1.000143
dtype: float64
```

Figure 10:Standardization data.

As shown above that mean become zero and standard deviation become 1.

❖ Exploratory Data Analysis (EDA)

- **Univariate Analysis**
  1- Visualize the distribution of key numerical features using **histograms.**

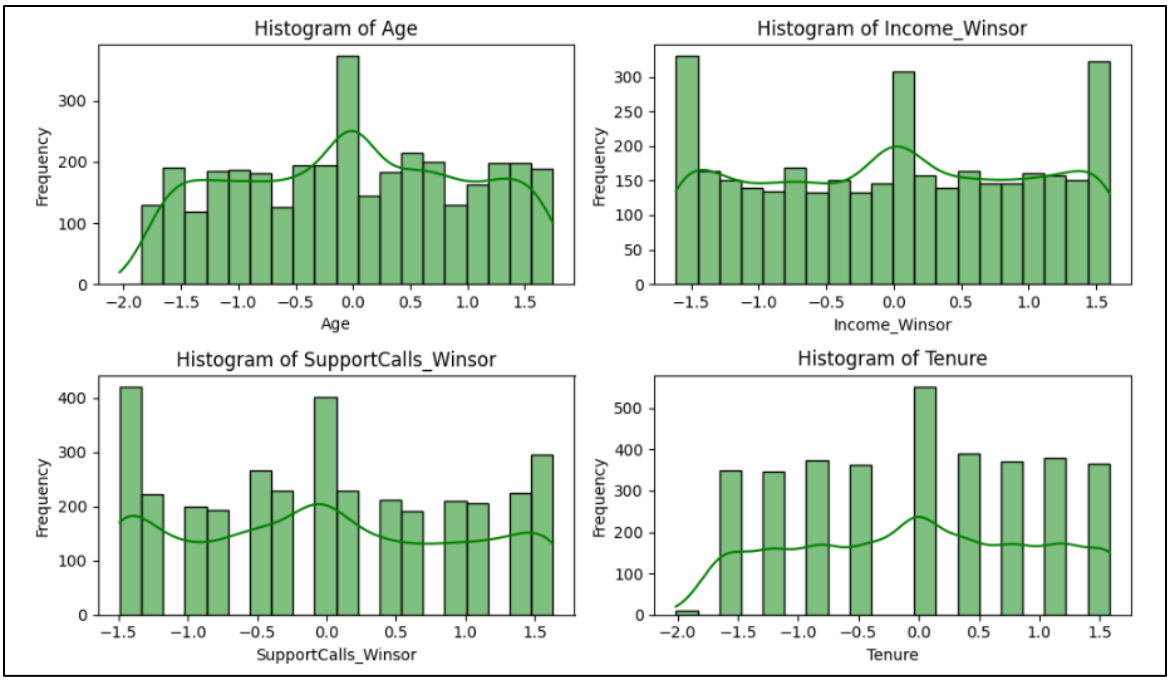


Figure 11:key numerical features using histograms

2- Analyze the distribution of categorical variables **using bar plots**.

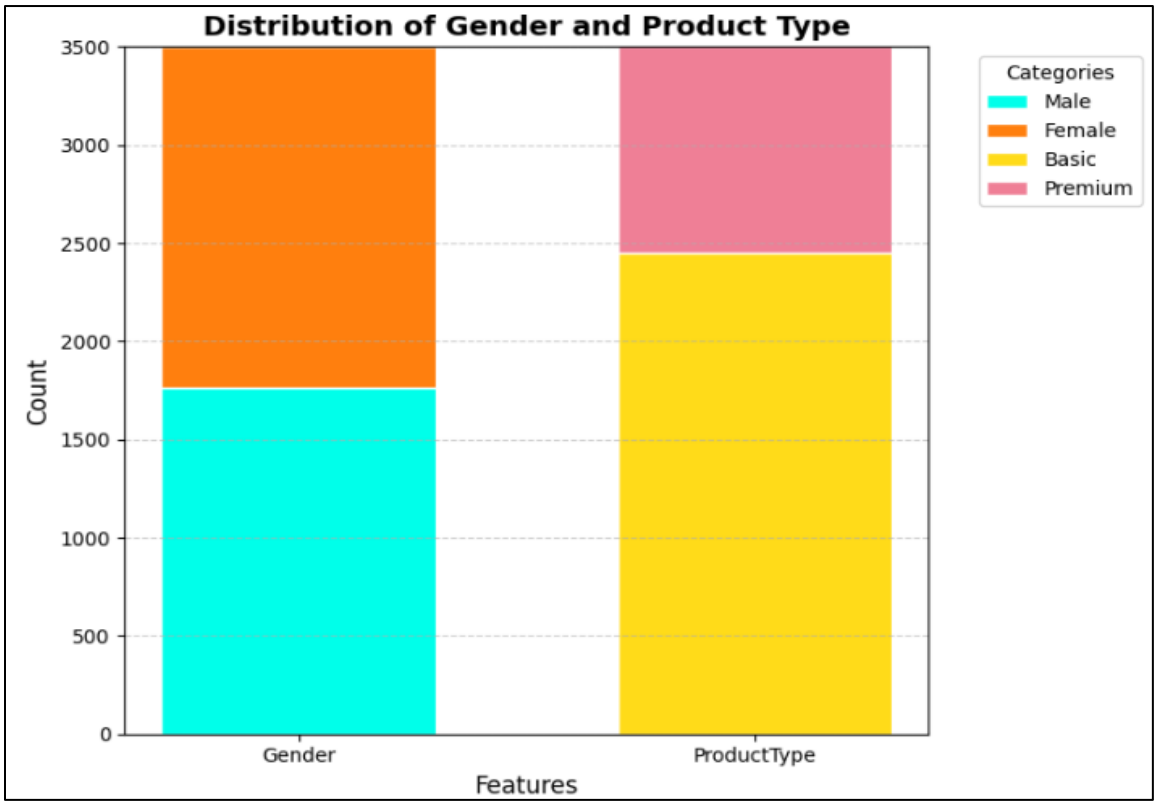


Figure 12:Distribution of categorical variables using bar plots.

- **Bivariate Analysis**

1- Explore relationships between numerical features and the target variable (ChurnStatus) using **box plots.**

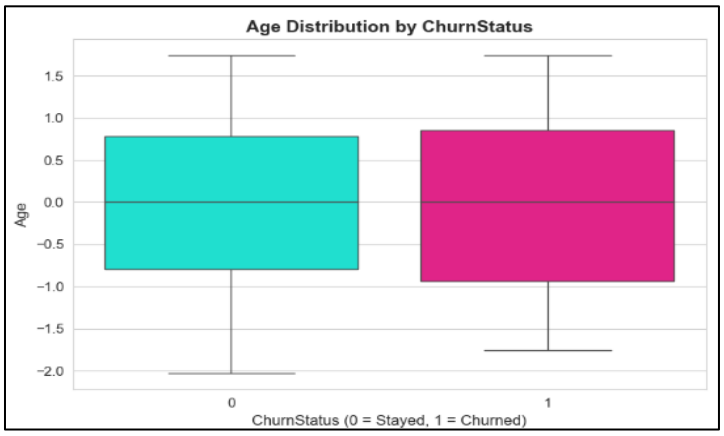


Figure 13:Age vs chuurnstate

As shown above in figure 13:

1-**Central Tendency Comparison (Median):** No strong relationship because the two median lines are roughly at the same level, the binary variable may not have a significant impact on the central value of the age.

2-Compare the Box Length (IQR): churned box is longer than stayed box it means that group churned has a wider spread or more variation in the numerical value.

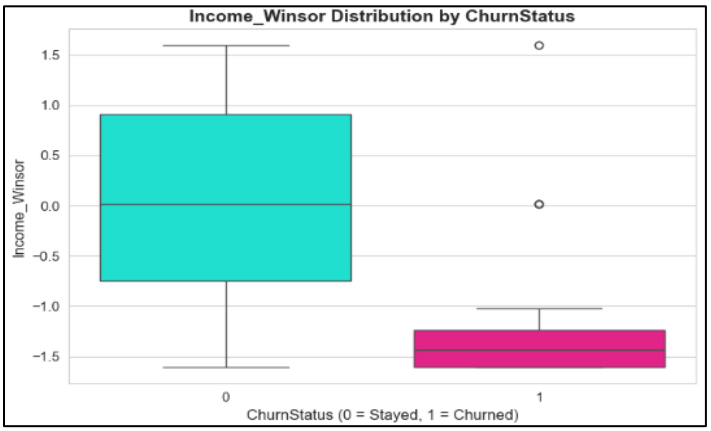


Figure 14:Income vs chuurnstate

As shown above in figure 14:

1-**Central Tendency Comparison (Median):**The median of stayed group is higher than churned group,it suggests a strong relationship and clearly associated with the numerical variable income.

2-Variability and Spread: stayed box is longer than churned box that mean stayed has highly consistent among the churned and much larger spread in income

3-Outliers: Churned has two outliers suggest that income is not the only factor influencing churn, as a few high-income individuals also left.

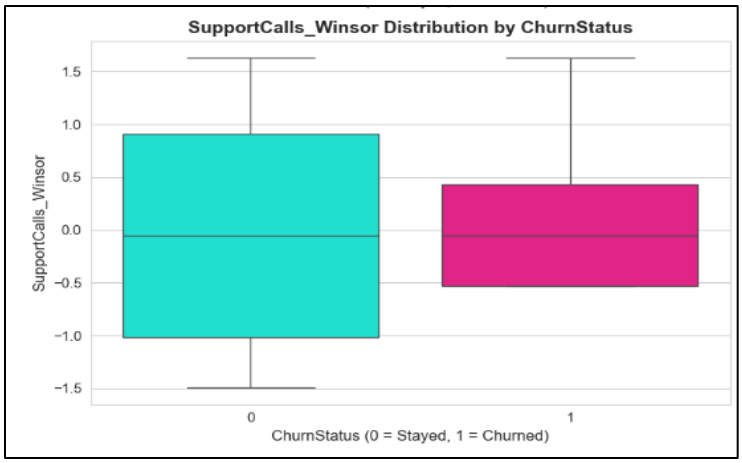


Figure 15:Supportcalls vs churned

As shown above in figure 15:

There is no strong relationship, stayed box has highly consistent among the churned and much larger spread in supportcalls values.

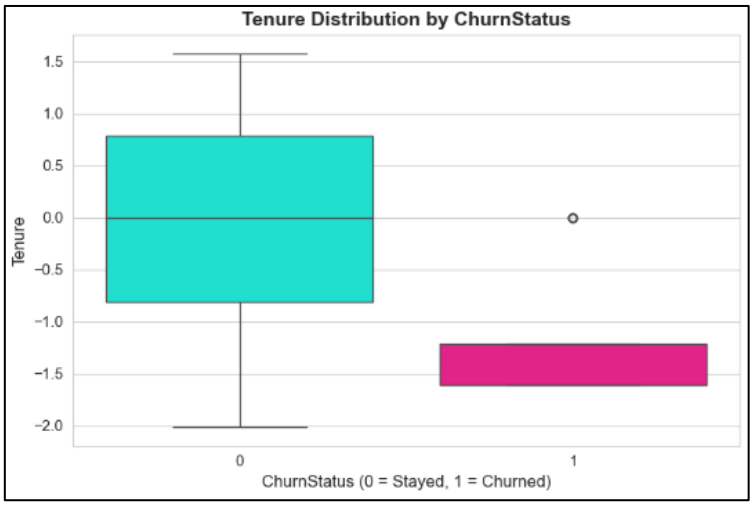


Figure 16:Tenure vs churned

As shown above in figure 16:

There is strong relationship, the stayed group exhibits a much wider spread of tenure values, while the churned group shows low tenure across the middle 50% of its members, with only one high-tenure outlier observed among the churners.

2- Investigate relationships between categorical variables and the target using **bar plots.**

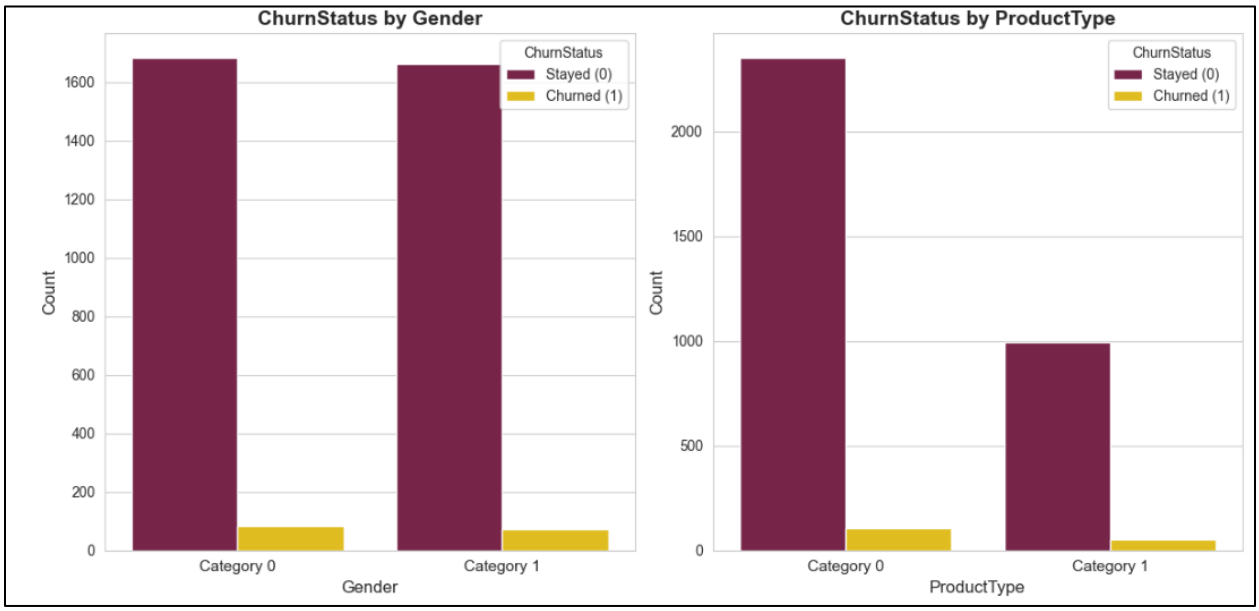


Figure 17:Churendstates vs categorical.

As shown above in figure17:

Gender: There doesn't appear to be a strong relationship between Gender and ChurnStatus.In other words, males and females churned and stayed at roughly the same rate.

ProductType: The Basic product (Category 0) has a larger total number of customers, but we can observe that Premium users (Category 1) may have a slightly lower churn count proportionally, so there may be a weak relationship between ProductType and ChurnStatus.Customers using the Premium product could be less likely to churn, possibly because they find more value in it.

- **Correlation Analysis**

As shown below in figure 18 :

Strength: As I noticed there is no strong correlation between features,but there is **very weak negative** correlation such as between Income and age with correlation of **-0.01**,there is **weak negative** correlation such as between Tenure and churnstatus with correlation of **-0.3**, there is **very weak positive** correlation such as between supportcalls and Age with correlation of **0.05** and there is **no correlation** at all such as between Age and producttype with correlation of 0.00.

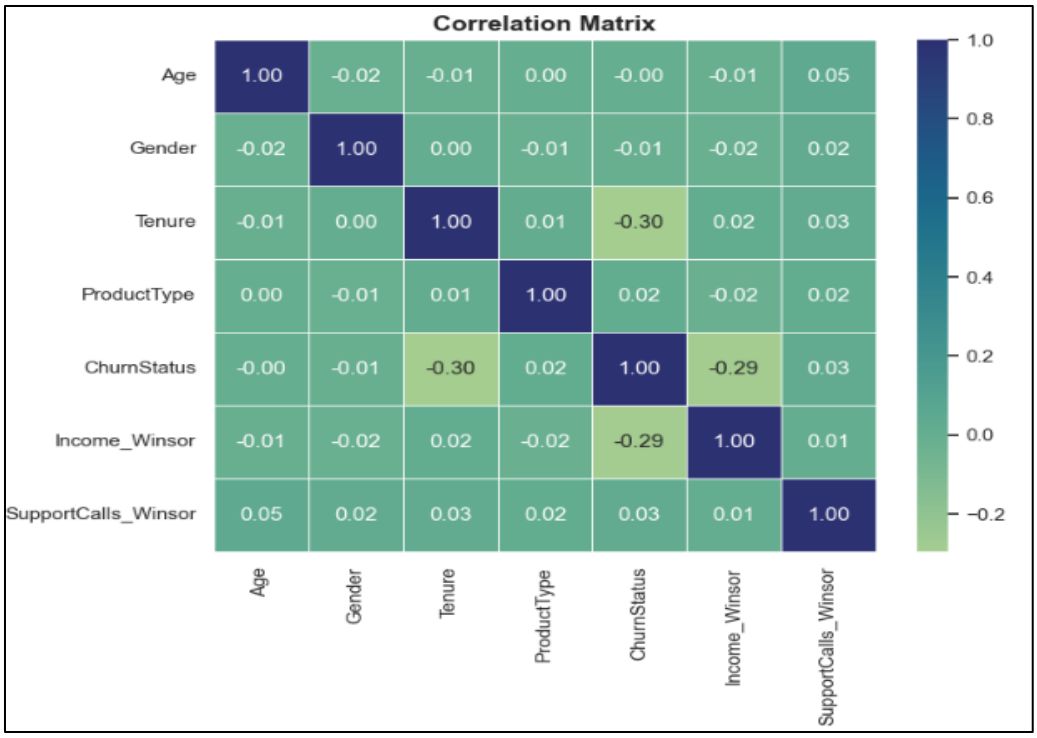


Figure 18:correlation matrix

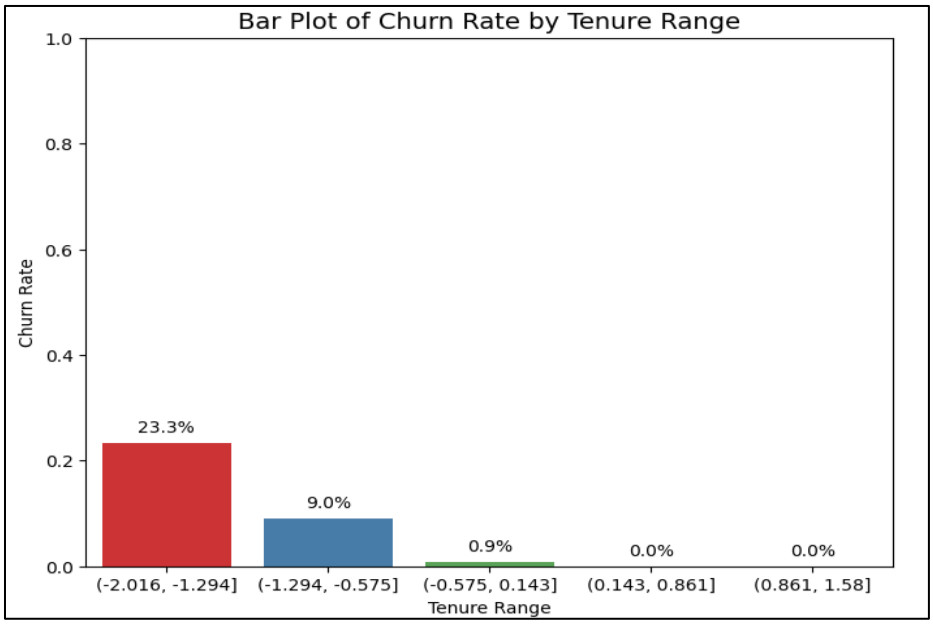
-Features seem most predictive of churn:

Tenure and Income are most likely predictive of churnstatues with negative correlation of -0.3, so as Tenure or Income increases, the likelihood of churn decreases. Correlation of -0.3 is weak, so these features have some predictive power, but not extremely strong.

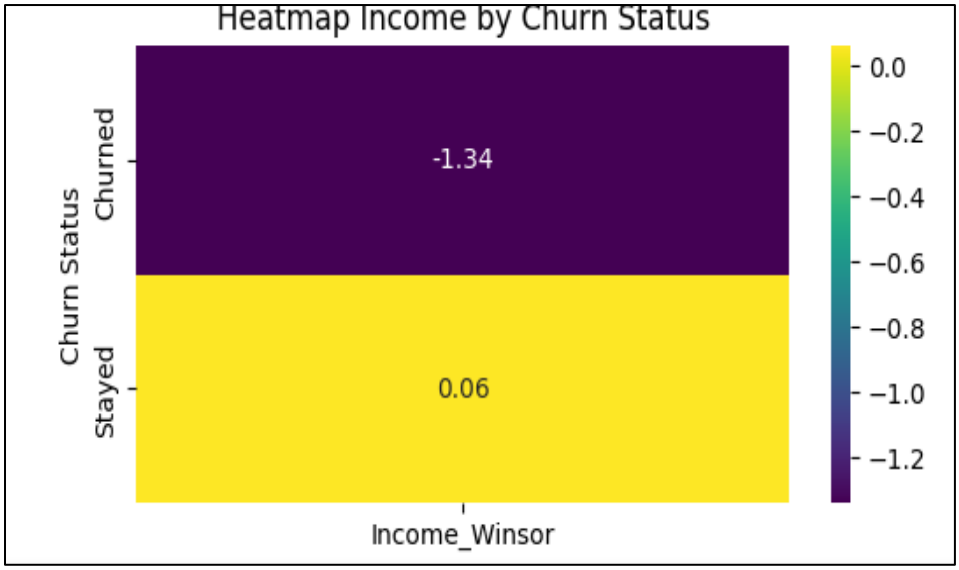
In other language: Customers who have been with the company longer (higher Tenure) or earn more (higher Income) are less likely to leave.

❖ Data Visualizations

I chose to visualize these data based on correlation matrix.



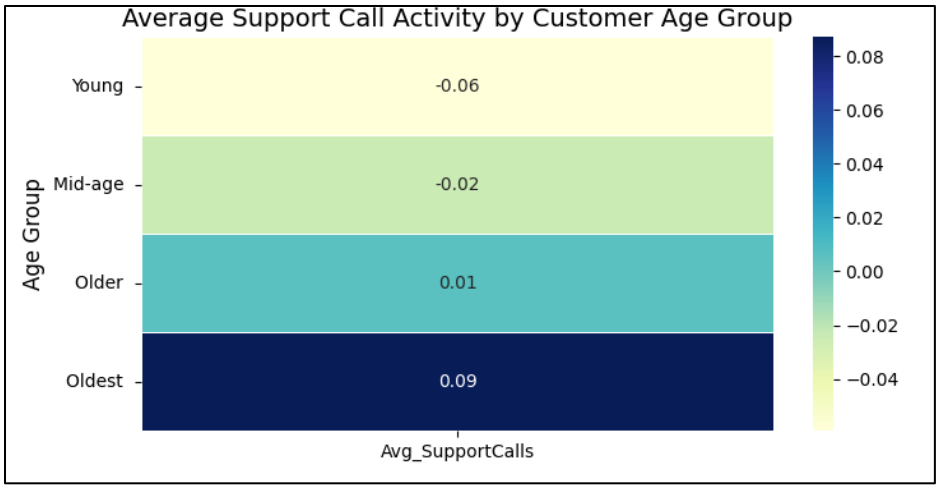
(a)



(b)

-Purple/Dark Box (Churned): The number -1.34 is the average income score of all customers who Churned.

-Yellow/Light Box (Stayed): The number 0.06 is the average income score of all customers who Stayed.



(c)

Every colorful box shows Customers in that group make an average number of support calls.

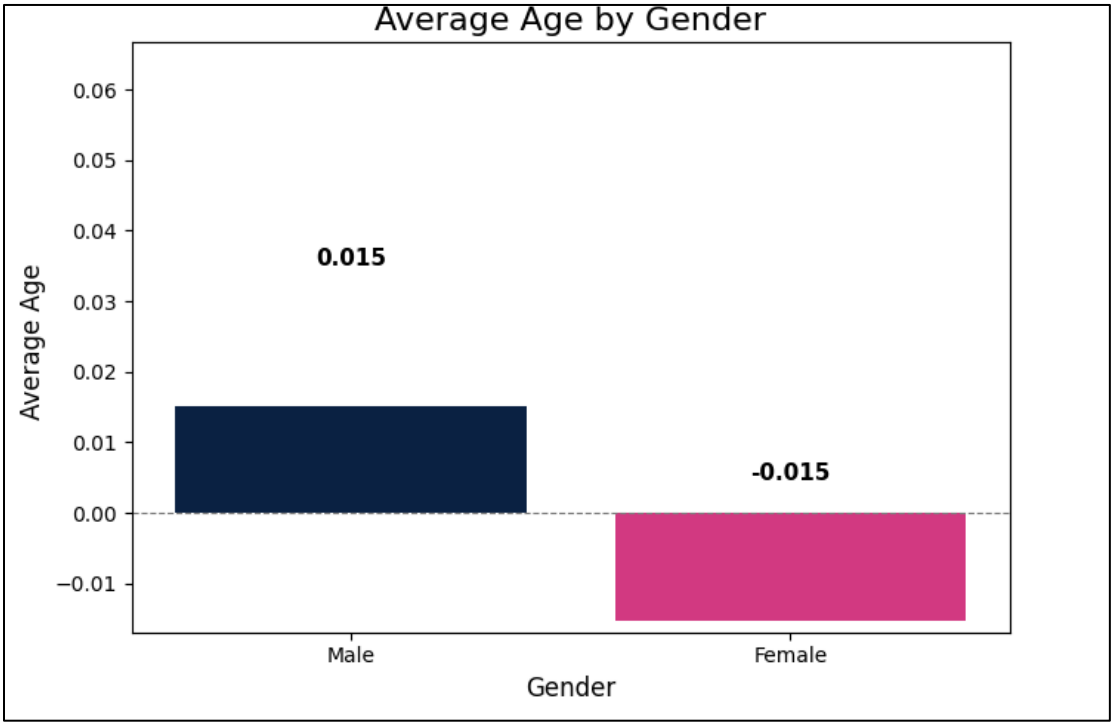


Figure 19: Visualizations of data.

A negative average score for a subgroup (like Female customers) means that their average raw age is slightly below the average age of the entire customer base.

❖ Conclusion

• Summarize the key findings from the EDA

Features	Correlation to churn
Tenure	-0.3
Income	-0.29
Age	-0.0
Gender	-0.01
ProductType	0.02
SupportCalls	0.03

• Discuss any significant patterns, relationships, or insights that could help improve customer retention or predict churn.

The analysis showed that Tenure and Income are the strongest predictors of churn. Customers with shorter tenures and lower incomes are more likely to leave the company. Therefore, a simple predictive model, such as Logistic Regression, should focus on these two features to effectively identify customers at risk of churning.

Variables like Age, Gender, Product Type and SupportCalls had minimal impact on churn prediction. Including them could introduce unnecessary noise and reduce model accuracy, so they should be excluded from the final predictive model.

THE END 😊